

Geography Ch-7 (Advanced) — The Thar Desert & Desertification in India

Advanced / India-applied | GS-I & GS-III (Environment) | Source: D.R. Khullar + Majid Husain

Thar (Great Indian Desert): extent, aridity, aeolian topography

Extra CA: ISRO-SAC Desertification & Land Degradation Atlas (distinct from Must-Do's Thar greening)

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

ADVANCED — The Thar (Great Indian Desert): Extent & Aridity (Khullar / Majid)

GS-I · India
Geography →
Arid Region
(Source: D.R.
Khullar, Majid
Husain)

■ NEWS TRIGGER

The Thar — India's only large hot desert — sits west of the Aravalli, covers ~1.42 lakh sq km (~4.3% of India), gets under 50 cm of rain, and is a live theatre of dunes, wind erosion and desertification.

■ INTRO & ORIGIN

Khullar places the arid/semi-arid Thar in Rajasthan and adjoining Punjab-Haryana, between the Indus and the Aravalli, mantled by sand that inhibits soil growth. The Rann of Kachchh is its extension.

*The sand derives from mechanical disintegration of local rocks (and blown-in material). West of the Aravalli — Jaisalmer and Barmer — is classic **aeolian topography** dominated by sand dunes.*

■ KEY DATA

Attribute	Value / detail	Source
Location	W of Aravalli; Rajasthan + Punjab/Haryana margins	Khullar
Area	~1.42 lakh sq km (~4.32% of India)	Khullar
Rainfall	< 50 cm (Khullar); < 25 cm core (Majid)	Khullar / Majid
Extension	Rann of Kachchh (Gujarat)	Khullar
Aeolian core	Jaisalmer, Barmer — sand dunes	Khullar
Crops / economy	Bajra, pulses, fodder; livestock; date-palm	Majid

■ STATIC THEORY

- The Thar lies **west of the Aravalli**, between the Indus and the Aravalli, spilling into Punjab, Haryana and northern Gujarat.
- Khullar: area **~1.42 lakh sq km (~4.32%)**, annual rainfall **< 50 cm**; a sand mantle inhibits soil formation. Majid's core figure is even drier (**< 25 cm**).
- The **Rann of Kachchh** is treated as an extension of the desert; **Jaisalmer and Barmer** show the strongest aeolian (dune) topography.
- Economy (Majid): **bajra, pulses, fodder**; livestock is central; scope for date-palm, water-melon and rain-water harvesting.
- It is described as the **Western Dry Region** — an arid tract (Marwar/Mewar margins) with extreme temperature range.

■ MUST-KNOW FACTS

1. Thar = west of Aravalli, between Indus & Aravalli.
2. Area ~1.42 lakh sq km (~4.32% of India); rain < 50 cm (Khullar).
3. Rann of Kachchh = extension of the Thar.
4. Jaisalmer & Barmer = strongest aeolian (dune) topography.
5. Core crops: bajra, pulses, fodder; livestock-based economy.

■ UPSC TRAPS

- ✗ The Thar lies east of the Aravalli.
- ✓ The Thar is WEST of the Aravalli range.
- ✗ The Rann of Kachchh is unrelated to the Thar.
- ✓ Khullar treats the Rann of Kachchh as an EXTENSION of the Thar desert.
- ✗ The Thar receives moderate rainfall (~75 cm).

✓ It receives < 50 cm (Khullar); the arid core is < 25 cm (Majid).

■ MAINS ANGLE

Characterise the Thar's extent, aridity and aeolian landscape, and link its physical setting to livelihood strategies and desertification risk.

■ STUDY LINK

Geography → India → Arid Region / Western Dry Region (Khullar; Majid: Physiography)

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA ANCHOR — ISRO-SAC Desertification & Land Degradation Atlas

GS-III • Arid geography applied → Land degradation policy

■ NEWS TRIGGER

ISRO's Space Applications Centre 'Desertification & Land Degradation Atlas of India' shows ~29.7% of India's land degraded (2018-19), with Rajasthan the largest desertified area — mostly by wind erosion.

■ INTRO & ORIGIN

Desertification is not just sand spreading — it is loss of dryland productivity by water/wind erosion and vegetation loss, mapped nationally to guide restoration.

■ STATIC THEORY

- ~29.7% of India (~97.85 mha) is degraded (2018-19), up from 28.76% (2003-05) — a rising trend.
- Rajasthan has the largest desertified area; **wind erosion** drives 40%+ of its desertification.
- India targets **Land Degradation Neutrality (LDN) by 2030** under the UNCCD; tools include Green India Mission and watershed/afforestation programmes.

■ MUST-KNOW FACTS

1. Atlas by ISRO-SAC; ~29.7% of India degraded (2018-19).
2. Rajasthan = largest desertified area (wind erosion dominant).
3. India target: Land Degradation Neutrality by 2030 (UNCCD).

■ UPSC TRAPS

- ✗ Desertification means only the spread of sand dunes.
- ✓ It means loss of dryland **PRODUCTIVITY** (water/wind erosion, vegetation loss) — often reversible.

■ MAINS ANGLE

Use the Atlas to argue for targeted, region-specific land-restoration to meet India's 2030 LDN goal.

■ STUDY LINK

Geography → Arid India → Desertification & LDN (applied CA)

End of Register Notes — UPSC Agent / Copilot CLI